FMA on the Litvak planted-error benchmark

100 planted errors over 10 paper(s). Reference run for $+X/-Y$: **FMA claude-opus-5**. Union of Litvak's systems: **93** of 100 – the empirical ceiling. *total flags* is every issue reported across the ten papers, matched or not: the column that separates a precise system from a verbose one. No system here, ours included, has a published precision figure.

1. Leaderboard

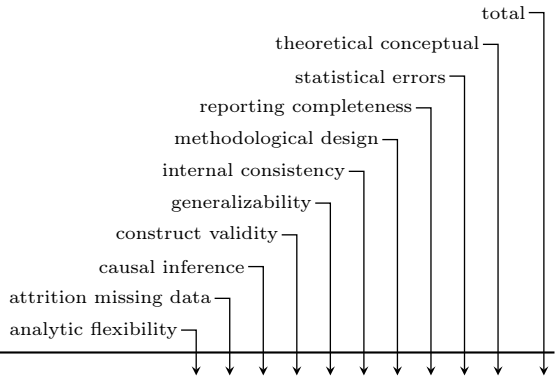
system	caught	total flags	$+X$	$-Y$
GPT-5.5 (high reasoning)	71	512	31	9
GPT-5.5 (no reasoning)	66	704	25	8
Claude Sonnet 4.6 (no thinking)	60	356	22	11
GPT-5.4 (no reasoning)	60	421	20	9
Claude Opus 4.6 (no thinking)	59	310	20	10
Refine.ink	57	305	18	10
GPT-5.4 (high reasoning)	56	278	18	11
Claude Opus 4.6 (high thinking)	55	293	18	12
Claude Opus 4.8 (high thinking)	52	227	11	8
Claude Sonnet 4.6 (high thinking)	51	241	17	15
FMA (Claude Opus 5)	49	206	0	0
Claude Opus 4.8 (no thinking)	47	279	9	11
Gemini 3.1 Pro	42	203	11	18
Gemini 3 Flash	36	200	12	25
Reviewer3	30	125	6	25

2. Adjusted (Litvak's correction: modified minus originals)

Nothing is planted in the unmodified papers, so every match there is the judge matching something to nothing. Across his 14 systems, $r(\text{findings}, \text{false positives}) = 0.726$: the raw leaderboard rewards volume. *not computed* = no clean-control run exists for that row.

system	raw	FP clean	adjusted	shift
GPT-5.5 (high reasoning)	71	24	47	+0
Refine.ink	57	19	38	+4
GPT-5.4 (no reasoning)	60	23	37	+1
Claude Opus 4.8 (high thinking)	52	16	36	+5
Claude Opus 4.6 (no thinking)	59	26	33	+0
GPT-5.5 (no reasoning)	66	34	32	-4
Claude Sonnet 4.6 (no thinking)	60	28	32	-4
Claude Opus 4.6 (high thinking)	55	26	29	+0
Claude Sonnet 4.6 (high thinking)	51	22	29	+1
Claude Opus 4.8 (no thinking)	47	20	27	+2
Gemini 3.1 Pro	42	17	25	+2
Reviewer3	30	9	21	+3
Gemini 3 Flash	36	16	20	+1
GPT-5.4 (high reasoning)	56	–	*n/c*	–
FMA (Claude Opus 5)	49	–	*n/c*	–

3. By category (10 planted errors each)



system										
GPT-5.5 (high reasoning)	8	7	6	7	7	6	9	7	7	7	71
GPT-5.5 (no reasoning)	7	7	4	7	7	8	8	6	7	5	66
Claude Sonnet 4.6 (no thinking)	7	7	4	6	6	4	8	6	5	7	60
GPT-5.4 (no reasoning)	5	6	4	6	5	4	8	8	7	7	60
Claude Opus 4.6 (no thinking)	5	6	5	5	4	9	8	5	6	6	59
Refine.ink	7	4	2	6	3	8	9	3	8	7	57
GPT-5.4 (high reasoning)	6	5	5	6	6	5	8	5	5	5	56
Claude Opus 4.6 (high thinking)	5	5	6	7	4	6	7	5	5	5	55
Claude Opus 4.8 (high thinking)	4	6	3	6	3	8	7	5	6	4	52
Claude Sonnet 4.6 (high thinking)	7	4	3	7	5	7	6	3	5	4	51
FMA (Claude Opus 5)	4	5	3	6	4	6	6	5	7	3	49
Claude Opus 4.8 (no thinking)	5	5	2	5	4	7	5	5	5	4	47
Gemini 3.1 Pro	4	6	3	6	3	5	4	3	6	2	42
Gemini 3 Flash	2	4	5	4	4	4	3	3	3	4	36
Reviewer3	3	6	4	3	2	2	3	2	4	1	30

Caveats

1. Litvak's figures are recomputed from his stored per-error results on the papers scored here, not quoted from his published leaderboard.
2. Different input: his systems read the flattened text PDF from 'docx_to_pdf.py'; ours read whatever the scored corpus was built from. A layout-preserving conversion favours us on table-bound checks.
3. Leaderboard and category figures use LITVAK'S PROTOCOL on every row: his haiku judge, one call per finding, no evidence requirement. That is how his systems were scored, so it is the only like-for-like comparison.

4. The ‘verified‘ column is a separate, stricter measurement of our runs only – one judgement per error, shown every finding for that paper, required to quote verbatim the words identifying the defect, with the quote checked in code. It is the better estimate of what we caught and is NOT comparable to the other rows.

5. ‘+X‘ = errors that system caught and the reference run missed. ‘-Y‘ = errors the reference run caught and it missed.

6. Litvak’s RAW scores are inflated in proportion to how much a system writes: r(findings, false positives on clean papers) = 0.726 over his 14 systems. His adjusted table is the defensible ranking, and we cannot yet appear on it – we have no clean-control run.